OSPF Operation

1.5.1

Video - OSPF Operation

Click Play in the figure to view a video about OSPF operation.

1.5.2

OSPF Operational States

Now that you know about the OSPF link-state packets, this topic explains how they work with OSPF-enabled routers. When an OSPF router is initially connected to a network, it attempts to:

- Create adjacencies with neighbors
- Exchange routing information
- Calculate the best routes
- Reach convergence

The table details the states OSPF progresses through while attempting to reach convergence:

State	Description
Down State	• No Hello packets received + Down. • Router sends Hello packets. • Transition to Init state.
Init State	• Hello packets are received from the neighbor. • They contain the Router ID of the sending router. • Transition to Two-Way state.
Two-Way State	• In this state, communication between the two routers is bidirectional. • On multiaccess links, the routers elect a DR and a BDR. • Transition to ExStart state.
ExStart State	On point-to-point networks, the two routers decide which router initiates the exchange of DBD packets and decide upon the initial DBD packet sequence number.
Exchange State	• Routers exchange DBD packets. • If additional router information is required then transition to Loading; otherwise, transition to the Full state.
Loading State	• LSRs and LSUs are used to gain additional route information. • Routes are processed using the SPF algorithm. • Transition to the Full state.
Full State	The link-state database of the router is fully synchronized.

1.5.3

Establish Neighbor Adjacencies

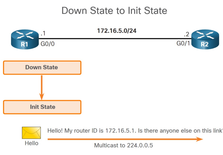
When OSPF is enabled on an interface, the router must determine if there is another OSPF neighbor on the link. To accomplish this, the router sends a Hello packet that contains its router ID not all OSPF-enabled interfaces. The Hello packet is sent to the reserved All OSPF Routers IPv4 multicast address 224.0.0.5. Only OSPFv2 routers will process these packets. The OSPF router ID is used by the OSPF process to uniquely identify each router in the OSPF area. A router ID is a 32-bit number formatted like an IPv4 address and assigned to uniquely identify a router among OSPF peers.

When a neighboring OSPF-enabled router receives a Hello packet with a router ID that is not within its neighbor list, the receiving router attempts to establish an adjacency with the initiating router.

Click each button below to step through the process routers use to establish adjacency on a multiaccess network.

1. Down State to Init State
2. The Init State
3. Two-Way State
4. Elect the DR and BDR

When OSPFv2 is enabled, the enabled Gigabit Ethernet 0/0 interface transitions from the Down state to the Init state. R1 starts sending Hello packets out all OSPF-enabled interfaces to discover OSPF neighbors to develop adjacencies with.



1.5.4

Synchronizing OSPF Databases

After the Two-Way state, routers transition to database synchronization states. While the Hello packet was used to establish neighbor adjacencies, the other four types of OSPF packets are used during the process of exchanging and synchronizing LSDBs. This is a three step process, as follows:

1. Decide first router
2. Exchange DBDs
3. Send an LSR

Click each button below to step through the process routers use to synchronize their LSDBs.

1. Decide First Router
2. Exchange DBDs
3. Send an LSR

1. Decide First Router

In the ExStart state, the two routers decide which router will send the DBD packets first. The router with the higher router ID will be the first router to send DBD packets during the Exchange state. In the figure, R2 has the higher router ID and sends its DBD packets first.

1.5.5

The Need for a DR

Why is a DR and BDR election necessary?

Multiaccess networks can create two challenges for OSPF regarding the flooding of LSAs, as follows:

- **Creation of multiple adjacencies** – Ethernet networks could potentially interconnect many OSPF routers over a common link. Creating adjacencies with every router is unnecessary and undesirable. It would lead to an excessive number of LSAs exchanged between routers on the same network.
- **Extensive flooding of LSAs** – Link-state routers flood their LSAs any time OSPF is initialized, or when there is a change in the topology. This flooding can become excessive.

To understand the problem with multiple adjacencies, we must study a formula:

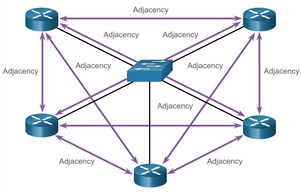
For any number of routers (designated as n) on a multiaccess network, there are $n \cdot (n - 1) / 2$ adjacencies.

For example, the figure shows a simple topology of five routers, all of which are attached to the same multiaccess Ethernet network. Without some type of mechanism to reduce the number of adjacencies, collectively these routers would form 10 adjacencies:

5 (5 – 1) / 2 = 10

This may not seem like much, but as routers are added to the network, the number of adjacencies increases dramatically. For example, a multiaccess network with 20 routers would create 190 adjacencies.

Creating Adjacencies With Every Neighbor



- Number of Adjacencies = n (n – 1) / 2
- n = number of routers
- Example: 5 (5 – 1) / 2 = 10 adjacencies

1.5.6

LSA Flooding With a DR

A dramatic increase in the number of routers also dramatically increases the number of LSAs exchanged between the routers. This flooding of LSAs significantly impacts the operation of OSPF.

Click each button to compare the flooding of LSAs without and with a DR.

Flooding LSAs | LSAs and DR

Flooding LSAs

To understand the problem of extensive flooding of LSAs, play the animation in the figure. In the animation, R2 sends out an LSA. This event triggers every other router to also send out an LSA. Not shown in the animation are the required acknowledgments sent for every LSA received. If every router in a multiaccess network had to flood and acknowledge all received LSAs to all other routers on that same multiaccess network, the network traffic would become quite chaotic.

1.5.7

Check Your Understanding - OSPF Operation

Check your understanding of OSPF operation by choosing the BEST answer to the following questions.

1. During this OSPF state on multiaccess networks, the routers elect a Designated Router (DR) and a Backup Designated Router (BDR).
   - Down State
   - Init State
   - **Two-Way State**
   - ExStart State
   - Exchange State
   - Loading State
   - Full State

2. During this OSPF state, routers send each other DBD packets.
   - Down State
   - Init State
   - Two-Way State
   - ExStart State
   - **Exchange State**
   - Loading State
   - Full State

3. An OSPF router enters this state when it has received a Hello packet from a neighbor, containing the sending Router ID.
   - Down State
   - **Init State**
   - Two-Way State
   - ExStart State
   - Exchange State
   - Loading State
   - Full State

4. During this OSPF state on point-to-point networks, the routers decide which router initiates the exchange of DBD packets.
   - Down State
   - Init State
   - Two-Way State
   - **ExStart State**
   - Exchange State
   - Loading State
   - Full State

5. During this OSPF state, routers have converged link state databases.
   - **Down State**
   - Init State
   - Two-Way State
   - ExStart State
   - Exchange State
   - Loading State
   - **Full State**

6. During this OSPF state, no Hello packets are received.
   - **Down State**
   - Init State
   - Two-Way State
   - ExStart State
   - Exchange State
   - Loading State
   - Full State

7. During this OSPF state, routes are processed using the SPF algorithm.
   - Down State
   - Init State
   - Two-Way State
   - ExStart State
   - Exchange State
   - Loading State
   - **Full State**

Check
Show Me
Reset